Math 1232 Summer 2025
Mastery Quiz 8
Due Tuesday, July 29

This mastery quiz has two topics. Everyone should do M3, but if you already have a 2/2 on S6, feel free to skip it. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write "yes" or "no" or give a single number.

Please turn in this quiz in class on Tuesday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

Topics on this quiz:

- Major Topic 3: Series Convergence
- Secondary Topic 6: Sequences and Series

Name:

Name: ______________________________

Major Topic 3: Series Convergence

Remember to show *all* your work—this includes stating any convergence tests you use, as well as computing any limits that arise.

(a) Analyze the convergence of the series $\displaystyle\sum_{n=2}^{\infty} \frac{n}{\ln(n)}$.

(b) Analyze the convergence of the series $\displaystyle\sum_{n=1}^{\infty} \frac{n}{n^4+1}$.

(c) Analyze the convergence of the series $\displaystyle\sum_{n=1}^{\infty} \frac{4n^3+1}{n^4+n+3}$.

Name: ____

Secondary Topic 6: Sequences and Series

(a) Compute $\displaystyle\sum_{n=1}^{\infty} \frac{2}{n^2+4n+3}$. Rigorously justify your computation of this limit.

(b) Consider the sequence $(a_n) = (6, 2, \frac{2}{3}, \frac{2}{9}, \dots)$. Find a formula for the nth term a_n. Compute $\lim_{n\to\infty} a_n$.

(c) Compute $\displaystyle\sum_{n=1}^{\infty} \frac{2^{2n+1}}{3^n}$.